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Master's Thesis

A Study on Effective Korean
Language Transfer Using Large
Language Models

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August 2024

A Study on Effective Korean Language Transfer Using Large Language Models

by

Seungjun Lee

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A thesis submitted in partial fulfillment of the
requirements for the degree of Master of Science

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A Study on Effective Korean Language Transfer Using Large Language Models

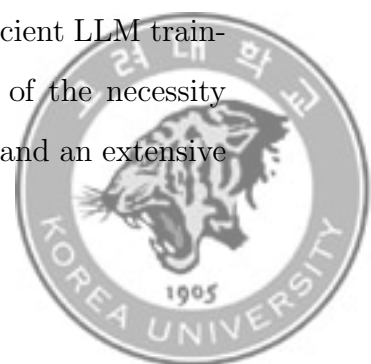
by Seungjun Lee

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Abstract

This paper presents a comprehensive study on the development and evaluation of Korean Large Language Models (LLMs) through effective language transfer and instruction-following methodologies. To overcome the limitations of Korean LLMs, we developed Korean LLMs by expanding the LLaMA2-7B base model using an interleaved block expansion approach, followed by post-training on a Korean corpus. We compared the performance of these models against other open-source Korean LLMs. Our findings indicate that our Korean LLM models outperform other open-source models. Additionally, we address the scarcity of Korean NLP data by constructing the KULLM dataset, which includes translated instruction-following datasets. Using this dataset, we trained our language-transferred models, resulting in a fine-tuned language model designed to accurately follow instructions in Korean. Key contributions of this study include the introduction of an efficient LLM training method through language transfer, the demonstration of the necessity and effectiveness of instruction-following models in Korean, and an extensive



performance comparison across various dataset combinations. Future work will focus on enhancing model performance using larger parameter models and more diverse datasets.

Keywords: Large Language Models (LLMs), Instruction-Following, Language Transfer, KULLM, Post-Training, Korean NLP



초거대 언어 모델을 활용한 효과적인 한국어 언어 전이 및 지시어 따르기 학습에 대한 연구

이 승 준

컴 퓨 터 학 과

지도교수: 임 희 석

초록

본 연구는 효과적인 언어 전이 및 지시어 따르기 (instruction-following) 방법론을 통해 한국어 초거대 언어 모델(LLM)의 개발 및 평가에 관한 종합적인 연구를 제안한다. 기존 한국어 LLM의 한계를 극복하기 위해, 우리는 LLaMA2-7B를 블록 확장 (block expansion) 접근법을 사용하여, 한국어 말뭉치에 대한 사후 학습을 통해 한국어 LLM을 개발했다. 해당 모델과 다른 오픈소스 한국어 LLM과 비교한 결과, 우리의 한국어 LLM 모델이 다른 오픈소스 모델보다 뛰어난 성능을 보임을 확인했다. 또한, 한국어 지시어 데이터의 부족 문제를 해결하기 위해 지시어 따르기 데이터셋을 번역하여 KULLM 데이터셋을 구축했다. 이 데이터셋을 사용하여 언어 전이 모델을 학습한 결과, 한국어 지시어를 정확하게 따르는 초거대 언어 모델을 만들 수 있었다. 본 연구의 주요 기여는 언어 전이를 통한 효율적인 LLM 학습 방법의 도입, 한국어 지시어 따르기 모델의 필요성과 효과를 입증, 다양한 데이터셋 조합에 대한 성능 비교를 통한 모델 성능 해석을 포함한다. 향후 연구에서는 더 큰 파라미터 모델과 다양한



데이터셋을 활용하여 모델 성능을 향상시키는 것을 목표로 한다.

주제어: 언어 전이, 초거대 언어 모델 (LLM), KULLM 데이터셋



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Chapter 1

Introduction

In recent years, the development of language models has led to remarkable advancements in the field of natural language processing (NLP). Large Language Models (LLMs), in particular, have significantly shifted the research landscape, demonstrating unprecedented capabilities in tasks ranging from text generation to comprehension. Following the success of early models like ChatGPT, cutting-edge research from OpenAI's GPT-4 and Meta AI's LLaMA has spearheaded major progress in this area, enabling a wide array of applications in both academic and industrial settings.

The impressive multilingual capabilities of LLMs stem from their pretraining on diverse corpora. However, the uneven distribution of language resources poses significant challenges in collecting extensive training data for all languages (Ranta and Goutte 2021). This disparity is evident in models like BLOOM, which, despite being pretrained on 46 languages, only covers a fraction of the world's approximately 7,000 languages, with substantial imbalances among them (Scao et al. 2023). Furthermore, in recent open-source LLMs like LLaMA, the amount of Korean training data is merely 0.1% of the entire pretraining data, highlighting the severe scarcity of Korean tokens in these models. This imbalance underscores the necessity for more focused efforts to improve the representation and performance



of low-resource languages like Korean.

Moreover, the success of LLMs such as ChatGPT and GPT-4 has highlighted the critical role of instruction-tuning in enhancing model performance (ChatGPT 2023). This tuning approach has also benefited open-source models like LLaMA, enabling them to achieve competitive performance at lower costs by focusing on high-quality, task-specific data (LLaMA 2023). These advancements underscore the need for targeted data collection and retraining efforts to improve LLM performance across a broader range of languages.

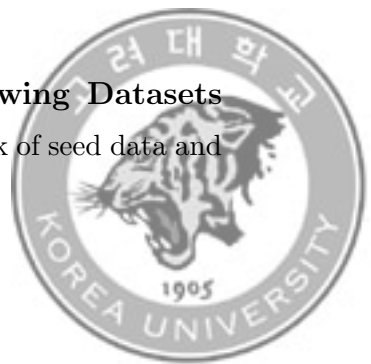
However, pre-training models inevitably leads to high computational and data collection costs, which are not feasible for low-resource languages. Additionally, most instruction-tuning research has been predominantly centered around English, underscoring the need for instruction-tuning and datasets for languages like Korean. Despite the release of models such as KoAlpaca and KoVicuna, there remains a lack of in-depth research on Korean instruction datasets and pre-trained models.

To address this gap, we introduce a method for training Korean pre-trained models through language transfer. Furthermore, we propose a method for instruction-tuning using a Korean dataset and conduct relevant experiments. The key contributions of this study are as follows:

Korean LLM Pre-training using Language Transfer To overcome the challenges of pre-training datasets for low-resource languages like Korean, we employ a specialized embedding initialization technique tailored for Korean. By leveraging a pre-trained model extensively trained on large-scale English corpora, we enable effective learning for the resource-scarce Korean language. This approach efficiently transfers knowledge from English to Korean, mitigating the data scarcity issue and enhancing the performance of the Korean LLM.

Methodology for Building Korean Instruction-following Datasets

The current Korean NLP ecosystem faces challenges due to a lack of seed data and



annotated datasets necessary for utilizing LLM outputs. We propose a comprehensive methodology for constructing instruction-tuning datasets in Korean, involving detailed steps for data collection, annotation, and integration. This methodology aims to facilitate effective instruction-tuning for Korean LLMs, ensuring the creation of high-quality, diverse, and representative datasets.

These contributions aim to advance the field of NLP for low-resource languages, particularly Korean, by addressing key challenges in data scarcity and instruction-tuning.



Chapter 2

Related Works

2.1 Korean LLM

Despite the advancements in LLMs and the release of numerous models and datasets by various organizations, there is a significant shortage of open-source Korean LLM models and datasets. Notable Korean LLMs include Polyglot-Ko [1] and the LLaMA-Ko¹ series. For instruction-following datasets, prominent examples are KoAlpaca² and KoVicuna³.

2.1.1 Pre-trained LLMs

Polyglot-Ko Polyglot-Ko is a Korean language model developed by EleutherAI to improve the performance of multilingual models in non-English languages. It was trained on 1.2TB of diverse Korean data sourced from blogs, news articles, patents, and Wikipedia, which was reduced to 863GB after preprocessing. The model is built upon EleutherAI's GPT-NeoX architecture.

¹<https://huggingface.co/beomi/llama-2-ko-7b>

²<https://github.com/Beomi/KoAlpaca>

³<https://github.com/melodysdreamj/KoVicuna>



LLaMA-2-Ko LLaMA-2-Ko⁴ is an optimized autoregressive language model based on LLaMA2 [2], utilizing the Transformer [?] architecture. It includes an expanded vocabulary and additional Korean corpora for further pretraining. By augmenting the original 32,000 vocabulary size of LLaMA2 with additional Korean vocabulary, the total vocabulary size was expanded to 46,336, thereby enhancing the model’s performance in Korean language tasks.

2.1.2 Instruction-following Datasets

KoAlpaca KoAlpaca is a model based on LLaMA and Polyglot-ko, trained on a Korean instruction dataset. The dataset for KoAlpaca was created by translating the instructions from the Alpaca [3] dataset using DeepL and then generating outputs with GPT-3.5-turbo [4].

KoVicuna KoVicuna is built on the LLaMA model and follows the approach of Vicuna [5]. This model was trained using a Korean-translated version of the ShareGPT⁵ dataset, which consists of conversational data between ChatGPT and users.

2.2 Post-training

2.2.1 Language Transfer

Post-training on corpora composed of languages different from those used in pretraining has shown promising results in adapting multilingual models to new languages. Alabi et al. (2022)[6] adapted XLM-R to up to 20 African languages simultaneously, rather than specializing in a single language. Ebrahimi and Kann

⁴<https://huggingface.co/beomi/llama-2-ko-7b>

⁵<http://sharegpt.com>



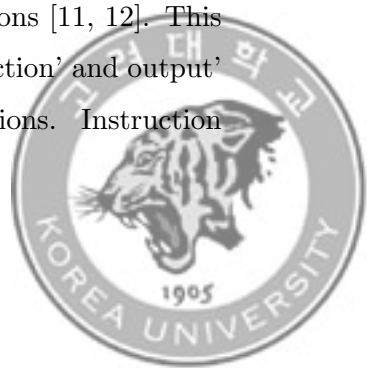
(2021)[7] and Wang et al. (2022)[8] utilized resources with broader language coverage than typical web-scraped monolingual texts, such as the Bible and lexicons, to adapt pretrained multilingual models to unseen languages. Muller et al. (2021)[9] improved results by transliterating unseen languages into the Latin script, thus leveraging existing pretrained vocabularies more effectively.

2.2.2 Block Expansion

In the context of advanced techniques for training large language models (LLMs), the post-pretraining method introduced by Wu et al. (2024)[10] in "LLaMA Pro: Progressive LLaMA with Block Expansion" is particularly noteworthy. This method, known as block expansion, addresses the issue of catastrophic forgetting, where an LLM loses previously learned information during new training. By expanding the model architecture with additional Transformer blocks fine-tuned on new domain-specific data while freezing the original blocks, LLaMA Pro significantly enhances the model's capabilities across diverse tasks without compromising its existing knowledge base. This approach not only preserves the general utilities of the base model but also systematically integrates new knowledge, extending the model's applicability and performance in specialized domains such as programming and mathematical reasoning.

2.3 Instruction Tuning

One of the key elements driving the advancement of LLMs is instruction tuning. Instruction tuning is a strategy aimed at enhancing the zero-shot learning capabilities of LLMs by fine-tuning the model with specific instructions [11, 12]. This process generally involves further training the LLM using instruction-output pairs to ensure the model can effectively follow given instructions. Instruction tuning is typically built using two methods [12]:



Output Generation Using LLMs This method involves using LLMs such as GPT-3.5-Turbo [4] or GPT-4 [13] to generate outputs for given instructions. Instructions can be collected in two ways: manually crafting them or creating a small initial seed of instructions, which are then expanded using the LLM. These expanded instructions are used as inputs to generate corresponding outputs from the LLM. Datasets like InstructWild [14] and Self-Instruct [15] utilize this approach.

Data Construction Using Annotated Datasets This approach uses templates to convert existing annotated natural language datasets into (instruction, output) pairs. Text-label pairs are transformed into instruction-output pairs, leveraging previously annotated data. Datasets like FLAN [16] and P3 [17] have been constructed using this method.



Chapter 3

Language Transfer

This chapter introduces methodologies for transferring large language models (LLMs) trained on a source language to a target language. In this study, we utilize LLaMA-2 [2], which is primarily trained on English, as the source language model, and the target language is set to Korean.

	Length	Content
Original Sentence	28	안녕하세요, 저는 한국어를 배우고 있는 학생입니다.
Original Tokenizer	41	‘_’, ‘안’, 0xEB, ‘0xEB’, 0x85, ‘0x85’, 0x95, ‘0x95’, ‘하’, ‘세’, ‘요’, ‘,’’, ‘_’, 0xEC, ‘0xEC’, 0xA0, ‘0xA0’, 0x80, ‘0x80’, ‘는’, ‘_’, ‘한’, ‘국’, ‘어’, ‘를’, ‘_’, 0xEB, ‘0xEB’, 0xB0, ‘0xB0’, 0xB0, ‘0xB0’, ‘우’, ‘고’, ‘_’, 0xEC, ‘0xEC’, 0x9E, ‘0x9E’, 0x88, ‘0x88’, ‘는’, ‘_’, ‘학’, 0xEC, ‘0xEC’, 0x83, ‘0x83’, 0x9D, ‘0x9D’, 0xEC, ‘0xEC’, 0x9E, ‘0x9E’, 0x85, ‘0x85’, ‘니’, ‘다’, ‘.’
Expanded Tokenizer	15	‘_안녕’, ‘하’, ‘세요’, ‘,’’, ‘_저’, ‘는’, ‘_한국어’, ‘를’, ‘_배우’, ‘고’, ‘_있’, ‘는’, ‘_학생’, ‘입니다’, ‘.’

Table 3.1: Tokenizer comparisons between original LLaMA and KULLM.



3.1 Korean Vocabulary Expansion

According to the study by Touvron et al. (2023) [18], the LLaMA training corpus consists of approximately 1.4 trillion tokens, primarily in English and a few European languages that use Latin or Cyrillic scripts. This composition equips LLaMA with strong multilingual and cross-lingual capabilities within the European language framework, but it limits its ability to understand and generate Korean text. To address this issue, we propose incorporating a Korean text corpus into LLaMA’s pre-training process to enhance its Korean comprehension and text generation capabilities. Our preliminary findings indicate that the existing model includes fewer than 1,000 Korean characters due to vocabulary limitations. Consequently, the model relies on byte-level tokenization for unknown UTF-8 characters, which increases sequence length and reduces processing efficiency.

3.1.1 Tokenizer Training

To address these challenges, Yang et al. (2022)[19] propose expanding the LLaMA vocabulary to include additional Korean tokens and adapting the model to accommodate these new vocabulary items. We initiated this expansion by developing a morpheme-aware subword tokenizer consisting of 20,000 words from a Korean corpus using SentencePiece [20]. For training the Korean morpheme-aware subword tokenizer, we sampled Korean corpora from various domains (e.g., Modu Corpus (모두의 말뭉치), AI Hub, Korean Wiki) as outlined in Appendix ?? . We pre-tokenized the 15GB sampled corpus using MeCab-Ko, then trained a BPE model with SentencePiece.

We merged this specialized tokenizer with the original LLaMA tokenizer to create an integrated tokenizer with a total of 49,953 tokens, termed the Korean LLaMA tokenizer. This integration preserved the integrity of the existing tokens while introducing new embeddings for the Korean tokens without altering the orig-



inal structure. Initial evaluations show that the Korean LLaMA tokenizer reduces token generation by almost half, effectively doubling encoding capacity and generation speed within the same context. These improvements significantly enhance LLaMA’s efficiency and capability in processing Korean text, as demonstrated in Table 3.1, indicating that the revised model successfully adapts to Korean language characteristics.

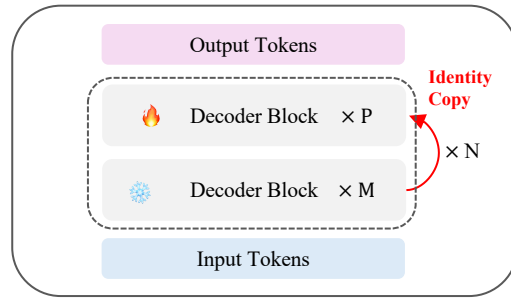
3.1.2 Morpheme-based Subword Embedding Initialization

When extending the vocabulary, the newly added vocabulary results in randomly initialized embeddings, which require significant resources during post-training. To address this, we propose a Morpheme-based Subword Embedding Initialization technique using the methodology of Wechsel [21]. The Subword Embedding Align method used in the original Wechsel shows low performance in languages with low token overlap, such as Chinese [22]. To improve this, we introduce morpheme subwords, leveraging the inherent morphological structure of such languages to achieve more meaningful and contextually relevant embeddings. We utilize morpheme-trained subword embeddings and morpheme-aware subword tokenization [20] for this purpose.

Morpheme Subword Embedding Morpheme subword embedding involves splitting subwords into morphemes during embedding calculations. We used FastText for this process, sampling Korean corpora from various domains (e.g., 모두의 말뭉치, AI Hub, Korean Wiki) as outlined in Appendix ?? . We pre-tokenized the 15GB sampled corpus using MeCab-Ko and then trained FastText. Using morpheme subword embeddings allows for more effective subword embeddings, as shown in Table 3.2.



Transformers Layers



Embedding

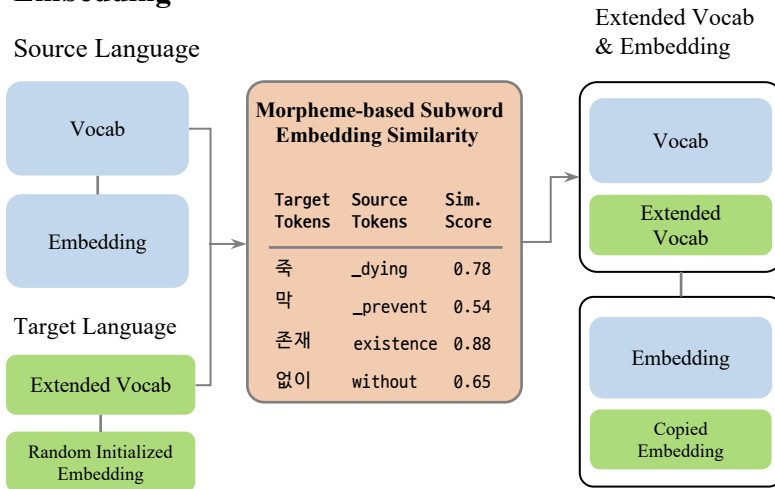


Figure 3.1: **Transformer Layers:** We employ backbone expansion and fine-tune the expanded identity blocks using the aspect corpus while freezing the blocks inherited from the base model. **Embedding:** We calculate morpheme-based subword embedding similarity for both source and target vocabularies (extended vocab) and align the embeddings for tokens with high similarity.



Word	Method	Top-5 Nearest Tokens
마시	w/ Morpheme	마신다, 마셔, 마십니다, 마셔서, 마셔도
	w/o Morpheme	코프와, 모턴, 매클레넌, 마히, 라버

Table 3.2: Top-5 nearest tokens for the root word ‘마시’ before and after applying morpheme-based subword embedding.

Morpheme-aware Subword Tokenization Morpheme-aware subword tokenization combines morphological analysis with subword tokenization. In this approach, we first use the MeCab-ko tokenizer to segment input sentences into morphemes. To address the issue of losing whitespace information during morpheme segmentation, we introduce a special token to represent spaces, allowing for the original sentence to be reconstructed. Then, we apply the BPE (Byte Pair Encoding) algorithm using SentencePiece to the list of morphemes.

This method ensures that BPE does not operate across morpheme boundaries. For example, in the sentence ‘나랑 쇼핑하자’, ‘쇼핑’ and ‘하’ are recognized as separate morphemes, preventing ‘핑’ and ‘하’ from being combined into a single subword. This approach leverages the linguistic properties of morphemes to generate more accurate and meaningful subword units. By maintaining morpheme boundaries while efficiently segmenting into subwords, this method performs well even with complex languages like Korean.

3.1.3 Morpheme-based Subword Similarity Transfer

We perform Morpheme-based Subword Similarity Transfer using morpheme subword embeddings and a morpheme-aware subword tokenizer.

First, we calculate subword embeddings for both the source and target languages within the same semantic space using morpheme subword embeddings. To achieve this, we decompose the subwords of the tokenizer vocabulary into n-grams and



compute the sum of the embeddings for all occurring n-grams. This method is similar to how fastText handles out-of-vocabulary word embeddings. If a subword contains unknown n-grams, the embedding is initialized to zero. By applying this process to both the source and target languages, we obtain subword embeddings for each vocabulary.

Next, we use these embeddings to perform an orthogonal Procrustes alignment, effectively mapping the source language subword embeddings to the target language. Using the resulting bilingual mapping matrix, we compute the cosine similarity between each target language subword and the source language subwords, as illustrated in Figure 3.1.

We then define each target language subword embedding as the weighted average of the k nearest source language embeddings, with weights calculated using the softmax of the similarities. For example, the target language token ‘죽’ is mapped to the nearest source language tokens ‘dying’, ‘dead’, and ‘death’ via the bilingual mapping matrix. The weighted average of the embeddings corresponding to these source language tokens is used as the embedding for ‘죽’. If a target language subword embedding is zero, it is randomly initialized based on the mean and variance of the source language embeddings.

3.2 Post-Training

We conduct post-training using embeddings initialized with a morpheme-based subword embedding initialization method. This process utilizes a collected Korean dataset. We employ the Block Expansion technique from LLaMA-Pro [10] to ensure that Korean language learning does not lead to forgetting the source language proficiency.



3.2.1 Block Expansion

Traditional language transfer methods often suffer from catastrophic forgetting during post-training, leading to a decline in performance on the original language. To address this, we leverage the findings from LLaMA-Pro, which focuses on domain adaptation without forgetting pre-learned knowledge. Our approach builds on this by ensuring that while learning a new language, the model retains knowledge of previously learned languages.

We expand the existing pre-trained LLM by adding additional Transformer blocks that are initially zero-initialized to enable identity mapping. These new blocks are fine-tuned with Korean corpora, while the original blocks remain unchanged to preserve the model’s prior knowledge.

Following the LLaMA-Pro configuration, we copy and insert eight layers into the existing layers. Unlike previous studies that freeze all layers except for the new ones, we do not freeze the embedding layers to learn the expanded embeddings through vocabulary expansion.

3.2.2 Post-Training Objective

We post-train the block-expanded LLaMA-2 model using the standard Causal Language Modeling (CLM) task. Given an input token sequence $\mathbf{x} = (x_0, x_1, x_2, \dots)$, the model predicts the next token x_i in an autoregressive manner. The objective is to minimize the following negative log-likelihood:

$$\mathcal{L}_{\text{CLM}}(\Theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\text{PT}}} \left[- \sum_i \log p(x_i \mid x_0, x_1, \dots, x_{i-1}; \Theta) \right] \quad (3.1)$$

where Θ represents the model parameters, $\mathcal{D}_{\text{PT}}$ is the post-training dataset (Korean Corpus), x_i is the token to be predicted, and $x_0, x_1, \dots, x_{i-1}$ constitute



the context.



Chapter 4

Building a Korean Instruction-Following Dataset

The current Korean NLP field lacks seed data for utilizing the outputs of LLMs and annotated datasets for constructing instruction-following datasets. This chapter proposes a methodology for instruction-tuning in Korean, which involves fine-tuning using instruction-following datasets.

4.1 KULLM Dataset

The KULLM (Korea University Large Language Model) dataset is a Korean instruction-tuning dataset designed for generating outputs using LLMs. Previous research, such as GPT-4-LLM [23], addressed the data scarcity issue for non-English languages by translating English instruction datasets into Chinese using GPT-4 for instruction-tuning. Similarly, in this study, we construct an instruction-following dataset by translating various instruction datasets and their outputs into Korean using DeepL¹. We chose DeepL because it has demonstrated superior translation performance compared to GPT-4 across multiple languages [24, 25].

¹<https://www.deepl.com/translator>



GPT-4-LLM [23] collected outputs for Alpaca’s instructions using GPT-4. This dataset consists of 520,000 instruction-output pairs, mirroring Alpaca’s dataset.

Dolly [26] is an instruction dataset created by Databricks. It includes 150,000 data points covering various behavior categories described in InstructGPT [27], such as brainstorming, classification, closed QA, generation, information extraction, open QA, and summarization.

ShareGPT is a dataset that collects conversations between ChatGPT and users, comprising 900,000 data points. In this study, we used user inputs as instructions and ChatGPT’s responses as outputs. We removed incorrectly translated English sentences using langid² and filtered out instructions and outputs with fewer than 10 characters.

Using GPT-4-LLM, Dolly, and ShareGPT, we constructed the **KULLM Dataset**³, a Korean instruction-following dataset containing a total of 1.5 million instruction-output pairs.

4.1.1 Supervised Fine-tuning

Pre-trained language models often fail to follow user instructions accurately and may generate unintended content. This occurs because the objective function of language modeling is next-token prediction, which differs from the objective of instruction-tuning, i.e., “follow the instruction and answer the question” [19]. To align the model’s behavior with user intent, we can explicitly fine-tune the model to follow instructions.

In this study, similar to Taori et al. (2023)[3], who used LLaMA, we train the instruction-following dataset named KULLM. The KULLM dataset is used for supervised fine-tuning (SFT). SFT is akin to causal language modeling, where the

²<https://github.com/saffsd/langid.py>

³<https://huggingface.co/datasets/nlpai-lab/kullm-v2>



model is prompted with an instruction and learns to generate outputs in an autoregressive manner. The instruction is included in a prompt template, followed by the output. For fine-tuning and inference, we adopt the following template used in Alpaca, with the input sequence as follows:

Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

Instruction: {instruction input}

Input: {input}

Response: {output}

The loss for SFT training is calculated only on the {output} part of the input sequence and can be expressed as follows:

$$\mathcal{L}_{\text{SFT}}(\Theta) = \mathbb{E}_{\mathbf{x}} \left[- \sum_{i \in \{\text{output}\}} \log p(x_i \mid x_0, x_1, \dots, x_{i-1}; \Theta) \right]$$

In this equation, given an input token sequence $x = \{x_0, x_1, \dots, x_{i-1}\}$, the model learns in an autoregressive manner to predict the next token x_i . Here, Θ represents the model parameters. Mathematically, the objective function aims to minimize the negative log-likelihood.



Chapter 5

Experiment

5.1 Experimental Setups

5.1.1 Experimental Setups for Post-Training

Data Preprocessing After collecting the Korean corpus described in Appendix ??, we performed preprocessing using the dps library. The preprocessing steps are as follows: i) We filtered out documents that were too long or too short by examining the length distribution of each dataset and setting appropriate length thresholds. ii) We filtered documents where the average word length did not fall within a specific range. iii) We filtered documents with an excessively high ratio of symbols (e.g., . ! ? @ # \$ % &) to regular words. iv) We filtered documents with a high proportion of lines starting with bullet points (e.g., *, ·, ●, ·,) or ending with ellipses (...). v) We filtered documents with too few alphabetic characters. vi) We reduced the use of excessive emoticons such as 'ㄱㄱㄱㄱ' and 'ㅇㅇㅇㅇㅇㅇ'. vii) We masked personal information such as phone numbers and social security numbers with tokens like <MASK>. viii) To ensure fast processing, we removed duplicate documents using Exact Match Deduplication.



Training Setups To create LLaMA-Pro-Ko-8B, we used LLaMA2-7B as the base model and expanded the number of blocks from 32 to 40 using an interleaved approach. During the block expansion process, we configured the parameters as $P = 1$, $M = 4$, and $N = 8$, resulting in 8 groups where each group expands from 4 blocks to 5 blocks. We freeze all blocks (layers) except for the additional blocks and embeddings. We performed post-training on the Korean corpus with a batch size of 960, a sequence length of 2048, a warmup ratio of 10%, a learning rate of 2×10^{-4} , and a cosine learning rate scheduler. We used bf16 mixed precision, a weight decay of 0.1, and gradient clipping at 1.0. To accelerate the training process, we applied the flash-attention mechanism. Our experiment was conducted on 8 NVIDIA A100 GPUs. LLaMA-Pro-Ko-8B was trained for a total of 8000 steps, corresponding to approximately 10 days.

Additionally, to demonstrate the effectiveness of the block expansion methodology by Wu et al. (2024) (**author?**) [10] in language transfer, we performed block expansion on the base model LLaMA2-7B under the same training conditions, followed by training all weights.

Evaluation Setups We evaluate the post-trained model on a total of 10 tasks, covering both English and Korean. The English evaluation is performed on five key general benchmarks using the Eleuther AI Language Model Evaluation Harness¹, a unified framework designed to test generative language models across a wide array of evaluation tasks. For Korean, we perform the evaluation using the Open Ko LLM Leaderboard², which operates in a leaderboard format.

The English benchmarks used for evaluation include:

- AI2 Reasoning Challenge (25-shot) (Clark et al., 2018) [28]: a set of grade-school science questions.

¹<https://github.com/EleutherAI/lm-evaluation-harness>

²<https://huggingface.co/spaces/upstage/open-ko-llm-leaderboard>



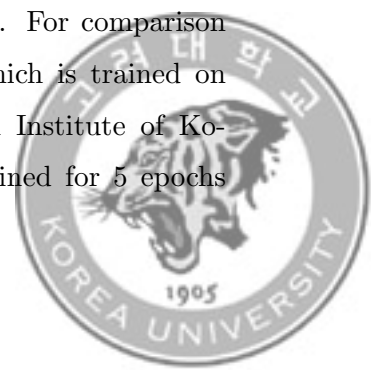
- HellaSwag (10-shot) (Zellers et al., 2019) [29]: a test of commonsense inference, which is easy for humans (approximately 95%) but challenging for SOTA models.
- MMLU (5-shot) (Hendrycks et al., 2020) [30]: a test to measure a text model’s multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.
- TruthfulQA (0-shot) (Lin et al., 2021) [31]: a test to measure a model’s propensity to reproduce falsehoods commonly found online.
- Winogrande (5-shot) (Sakaguchi et al., 2021) [32]: an adversarial and difficult Winograd benchmark at scale, for commonsense reasoning.

The Korean Open Ko LLM leaderboard includes the following evaluations:

- Ko-HellaSwag [33]: Translated from HellaSwag using machine translation.
- Ko-MMLU [34]: Human translation and variation of MMLU.
- Ko-ARC [35]: Human translation and variation of the ARC Challenge.
- Ko-Truthful QA [36]: Human translation and variation of TruthfulQA.
- Ko-CommonGen V2 [37]: Created from scratch, this benchmark uses Korean CommonGen to evaluate Korean language capabilities.

5.1.2 Experimental Setups for Instruction Fine-Tuning

Training Setups In this experiment, we train the LLaMA-Pro-Ko model, post-trained as described in Section 4.1, using the KULLM dataset. For comparison with a model trained from scratch, we select Polyglot-Ko, which is trained on 863GB of diverse Korean data, including blogs, the National Institute of Korean Language corpus, and Wikipedia [1]. All models are trained for 5 epochs



using four A100 GPUs (80GB VRAM each). We employ parameter-efficient training with LoRA using the PEFT³ library and optimize memory efficiency during training with bitsandbytes⁴. The AdamW optimizer [38] is used with a peak learning rate of 3×10^{-4} and a 5% warm-up cosine scheduler. Gradient clipping is applied to mitigate gradient explosion. We use a batch size of 256, selected for its superior performance among {128, 256, 512}, with LoRA Rank and Alpha set to 16 and 32, respectively. The decoding settings are temperature=0.1, top_k=40, top_p=0.75, do_sample=False, num_beams=4, repetition_penalty=1.4, and max_new_tokens=400.

5.1.3 Evaluation Setups

This section describes the evaluation methodology for assessing the instruction-following capability of models trained on the KULLM dataset in Korean. Evaluating LLMs is a complex and challenging task due to the lack of standards for accurately measuring the diversity, ambiguity of instructions, and the extensive capabilities of the models [39]. To address these challenges, several studies have employed the use of LLMs to evaluate other LLMs, demonstrating high correlation with human annotations across various benchmarks [39, 40].

Consequently, we adopt an evaluation approach centered around prompts described in natural language to assess the capabilities of the model trained on the KULLM dataset as well as other Korean LLMs. We designed prompts that implement a unified evaluation schema to comprehensively understand the output quality of a single model. This schema is based on the method proposed by LLM-Eval [40] and includes evaluation aspects such as content, grammar, relevance, and appropriateness. In our study, we set the evaluation metrics to include comprehension, naturalness, context maintenance, interest, accurate reflection of instructions,

³<https://github.com/huggingface/peft>

⁴<https://github.com/TimDettmers/bitsandbytes>



and overall output quality. The detailed prompts are as follows.

You are provided with a conversation between two people. You will receive the following instruction, input, and a corresponding response.

Your task is to evaluate the response based on the evaluation criteria outlined below.

It is important to read and understand these criteria thoroughly. Keep this document open for reference during the evaluation process.

Evaluation Criteria:

- Understanding (0 - 1): Can you understand the response based on the input?*
- Naturalness (1 - 3): Does the instruction sound like something a person would naturally say?*
- Context Maintenance (1 - 3): Does the response maintain context considering the input?*
- Interesting (1 - 3): Is the response boring or interesting?*
- Instruction (0 - 1): Is the response generated based on the instruction?*
- Overall Quality (1 - 5): What is your overall impression of the response's quality based on the above criteria?*

Evaluation Steps:

- 1. Carefully read the instruction, input, and response.*
- 2. Evaluate the response based on the criteria above.*

Instruction: {instruction}

Input: {input}

Response: {response}

In this study, we evaluated the performance of Large Language Models (LLMs)



Model	English					Korean					
	AC	HS	MM	TQ	WG	AC	HS	MM	TQA	KCG	AVG
LLaMA-2	53.07	78.59	46.87	38.76	74.03	31.91	41.68	34.11	38.49	30.34	46.78
LLaMA-2-Ko	48.46	75.28	39.56	34.49	72.14	38.05	49.58	30.48	37.06	41.56	46.66
LLaMA-2-Ko-8B	49.84	75.48	39.86	34.35	72.30	39.02	50.27	35.30	38.67	42.15	47.72
LLaMA-Pro-Ko-8B	53.24	77.93	47.06	38.32	72.22	40.19	51.26	36.80	40.24	43.80	50.11

Table 5.1: Results of post-training experiments in English and Korean. The abbreviations are as follows: AC: ARC Challenge, HS: HellaSwag, MM: MMLU, TQ: TruthfulQA, WG: Winogrande, KCG: Ko-CommonGen V2.

using GPT-4. The dataset used for evaluation was a Korean translation of the Human Eval dataset from Self-instruct⁵. This dataset was constructed to assess the practical performance of instruction-following models, extending beyond typical NLP tasks to include user-generated instructions for real-world applications. For example, when a user provides an instruction such as “Create a technical assessment questionnaire for R (programming language),” the model generates an appropriate response based on that instruction. Ultimately, the Instruction and Input fields of the evaluation prompts contained the translated Human Eval data, while the Response field contained the model-generated outputs.

5.2 Results

5.2.1 Results on Post-Training

We conducted comparative experiments based on the LLaMA model to measure the performance of language transfer through post-training. LLaMA-2-Ko is a model post-trained on LLaMA-2 in Korean. LLaMA-2-Ko-8B applies LLaMA-Pro’s Block Expansion to LLaMA-2, followed by training all layers. LLaMA-Pro-

⁵https://github.com/yizhongw/self-instruct/tree/main/human_eval



	Base Model	Fine-tuned	Understanding	Natural	Context	Interesting	Instruction	Overall
<i>closed</i>	GPT-3.5	-	0.980	2.806	2.849	2.056	0.917	3.905
	GPT-4	-	0.984	2.897	2.944	2.143	0.968	4.083
<i>open</i>	Polyglot-Ko _{12.8b}	KoAlapca	0.651	1.909	1.901	1.583	0.385	2.575
	Polyglot-Ko _{5.8b}	KULLM	0.754	2.099	2.187	1.746	0.579	3.028
	Polyglot-Ko _{12.8b}	KULLM	0.802	2.480	2.310	1.996	0.754	3.325
	LLaMA _{7b}	KoVicuna	0.460	1.583	1.726	1.528	0.409	2.440
	LLaMA-2 _{7b}	KULLM	0.723	1.874	1.923	1.552	0.450	2.993
	LLaMA-Pro-Ko _{8b}	KULLM	0.852	2.690	2.455	2.011	0.889	3.784

Table 5.2: Performance comparison experiments between fine-tuned on KULLM dataset. Understanding denotes the comprehensibility of the instruction, Natural refers to the naturalness of the response, Context is for context maintenance, Interesting indicates the level of interest in the response, Instruction denotes how well the instruction was followed, and Overall indicates the overall quality. Open and closed denote whether the model is open source or not.

Ko-8B, on the other hand, trains the expanded layers while freezing the rest.

Table 5.1 compares the performance of various LLaMA models on English and Korean evaluations. LLaMA-2-Ko shows improved performance in Korean due to post-training, but its English performance declines compared to LLaMA-2. Similarly, LLaMA-2-Ko-8B demonstrates higher performance in Korean tasks compared to LLaMA-2-Ko, yet lower performance in English. This indicates that post-training in Korean can cause catastrophic forgetting, reducing existing English capabilities (knowledge and reasoning). In contrast, LLaMA-Pro-Ko-8B exhibits minimal reduction in English performance while significantly enhancing Korean performance. This suggests that LLaMA-Pro’s block expansion methodology is effective for language adaptation.

5.2.2 Results on Instruction Fine-Tuning

This section presents the experimental results comparing the instruction-following performance of the model fine-tuned with the KULLM dataset against various Ko-



rean LLM models. Table 5.2 shows the instruction-following performance of the models trained on the KULLM dataset and other available Korean LLM models. In this experiment, the GPT-4 model achieved the highest performance across all evaluation metrics. Specifically, GPT-4 scored 4.083 in overall quality, significantly outperforming the other models.

Among the open-source Korean models, the LLaMA-Pro-Ko_{8b} model demonstrated the highest performance. This model excelled in comprehensibility, naturalness, context maintenance, interest, instruction adherence, and overall quality. Notably, it scored 3.784 in overall quality, significantly surpassing both KoAlpaca_{12.8b} and KoVicuna_{7b}.

The Polyglot-Ko_{5.8b} model, fine-tuned with the KULLM dataset, showed slightly lower performance compared to Polyglot-Ko_{12.8b} but still outperformed KoAlpaca_{12.8b} and KoVicuna_{7b}. The experimental results indicate that while the LLaMA-Pro-Ko_{8b} model does not yet match the performance of the GPT-4 model, it demonstrates superior performance among the open-source Korean models.

Ablation Study The model trained on the KULLM dataset demonstrated superior performance compared to other open-source Korean models. This performance difference can be attributed to various factors such as the training dataset, training code, and the libraries used. A key point of discussion is the performance gap observed in Table 5.2 between models trained on KULLM and KoAlpaca, despite both using the same Polyglot-Ko architecture. A closer examination reveals that while KoAlpaca was trained on a dataset of 520,000 instructions, the model trained on the KULLM dataset utilized a diverse set of instruction datasets. This difference in dataset composition and size likely has a significant impact on model performance. Therefore, we conducted an ablation study to investigate the impact of dataset composition on model performance.

For the ablation study, we examined the impact of individual datasets—GPT-4-LLM, Dolly, and ShareGPT—and their combinations on model performance.



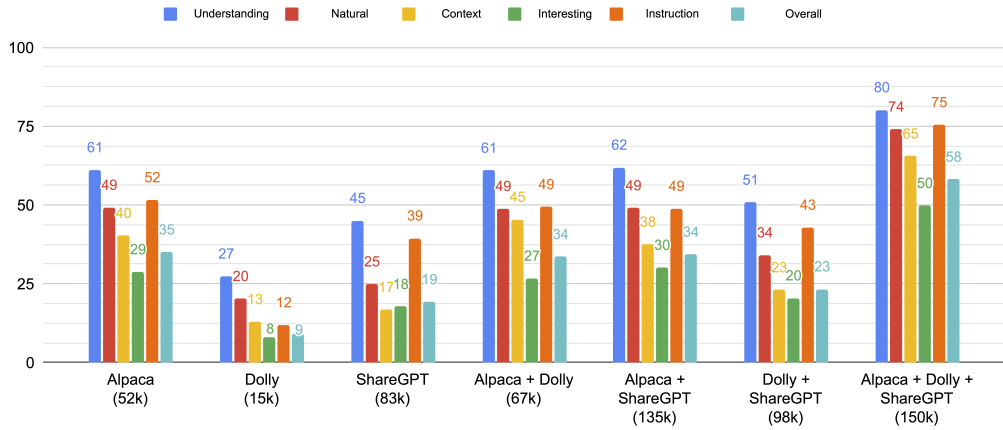


Figure 5.1: Performance comparison of instruction-following models based on combinations of GPT-4-LLM, Dolly, and ShareGPT datasets. All scores are rescaled to a range of 0-100.

Based on the results presented in Figure 5.1, we can draw the following key findings:

- Performance of Individual Datasets:** The Alpaca dataset, with a size of 52k, achieved the highest performance among individual datasets. Despite having 830k entries, the ShareGPT dataset showed lower performance than Alpaca. Dolly, the smallest dataset with 150k entries, exhibited the lowest performance.
- Dataset Combinations:** The combination of Alpaca, Dolly, and ShareGPT datasets, totaling 1.5 million entries, resulted in the highest performance. This suggests that each dataset provides complementary information, enhancing the model's overall capabilities. Notably, the combined datasets outperformed any single dataset significantly.
- Relationship Between Dataset Size and Performance:** There was a general trend that larger datasets led to better performance. However, performance was not solely determined by size. For instance, the combination



of Alpaca and ShareGPT, with 1.35 million entries, performed similarly to the Dolly and ShareGPT combination, despite the former being larger.

These results indicate that both the combination and size of datasets significantly impact model performance. Specifically, diverse dataset combinations substantially improve the model’s performance.



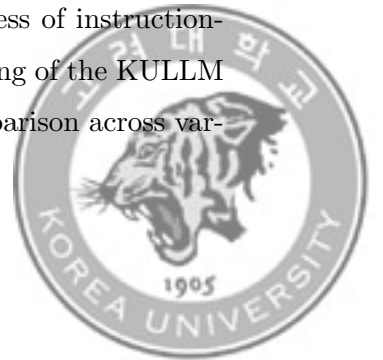
Chapter 6

Conclusion

In this study, we developed a Korean LLM through language transfer, constructed an instruction-following dataset suitable for Korean NLP, and trained the KULLM model based on these instructions. To address the scarcity of Korean data and resource limitations, we proposed an efficient post-training methodology for LLMs via language transfer. Recognizing the lack of instruction datasets in the Korean NLP ecosystem, we translated English instruction datasets into Korean, successfully creating the KULLM dataset.

To overcome the limitations of pre-trained language models, we applied supervised fine-tuning (SFT) during the training of the KULLM model. This approach allowed us to train a language model that accurately follows instructions. Evaluation results showed that the KULLM model outperformed other open-source Korean models. However, there remains a performance gap compared to commercial models like GPT-4.

The key contributions of this study are threefold. First, we introduced an effective LLM training method for target languages through efficient language transfer. Second, we demonstrated the necessity and effectiveness of instruction-following language models created for Korean through the training of the KULLM model. Third, we performed a comprehensive performance comparison across var-



ious dataset combinations to interpret the performance of the KULLM model.

However, due to resource constraints, the language transfer approach did not demonstrate effectiveness in larger parameter models. Additionally, the SFT method had limitations in building instruction datasets with annotations, such as FLAN, and in evaluating various NLP tasks. Future research will aim to enhance the performance of the KULLM model by utilizing models with different parameters and datasets.



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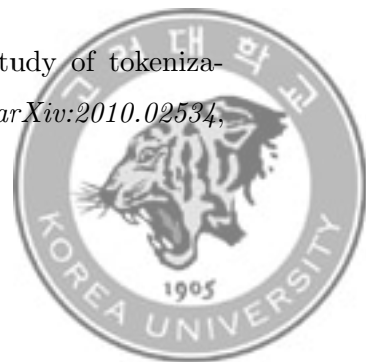
[4]



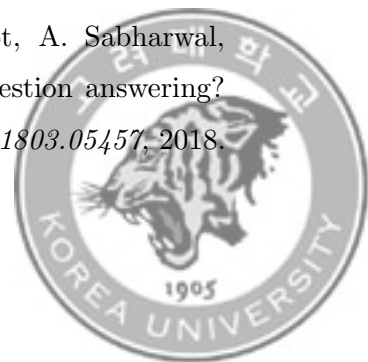
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